

Nirma University
Institute of Technology
Semester End Examination (RPR), May - 2024
Master of Computer Applications, Semester - I
6CS103 Operating System

Roll /
Exam No.

Supervisor's initial
with date

Time: 3 Hours

Max. Marks: 100

Instructions:

1. Attempt all questions.
2. Figures to the right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Assume suitable data wherever necessary and mention the same.
5. Section wise separate answer book should be used.

SECTION-I

Q-1. Answer the following. [16]

(A) **CO-1** What is an interrupt? How are simple and multiple interrupts (8)
BL-2 dealt with? Explain in detail.

(B) **CO-1** List and briefly explain five storage management responsibilities of (8)
BL-2 a typical Operating system.

Q-2. Answer the following. [16]

(A) **CO-2** Design and discuss a seven-state process model using a proper (8)
BL-2 state transition diagram and explain possible states and transitions.

(B) **CO-2** What is Thread? How it is different than a process? What is the (8)
BL-3 concept of multithreading? Is it the same as multiprocessing? Discuss.

Q-3. Answer the following. [18]

(A) **CO-3** Write the Solution to the Readers/Writers Problem Using (8)
BL-3 Semaphore where 'Writers' have priority

(B) **CO-3** Consider the following snapshot of a system (10)
BL-5

Process Name	Allocated				Maximum Demand				Available Matrix			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	3	0	1	1	4	1	1	1	2	0	2	0
P1	0	1	0	0	0	2	1	2				
P2	1	1	1	0	4	2	1	0				
P3	1	1	0	1	1	1	1	1				
P4	0	0	0	0	2	1	1	0				

Answer the following question:

1. What is the content of the need Matrix?
2. Is the system in a safe state? If yes, give a safe sequence else

justify.

3. If a request comes from P1 of (0,0,1,0) can we immediately grant the request? If yes, then check whether the system is in a safe state.

SECTION-II

Q-4. Answer the following. [16]

- (A) **CO-3** Define deadlock. How does it differ from Livelock? List and explain (8)
BL-2 conditions for deadlock in a system.
 (B) **CO-2** How do external fragmentation and internal fragmentation differ? (8)
BL-2 Explain the difference between both of them with an example.

Q-5. Answer the following. [16]

- (A) **CO-2** Consider a process that has 8 virtual pages and is assigned a fixed (8)
BL-5 allocation of three page frames in main memory. The pages are referenced in the following order.

0, 3, 2, 1, 4, 5, 3, 1, 6, 7, 2, 4, 6, 5, 1, 3, 2, 0, 3, 7, 5, 3, 1, 2.

If the main memory frames are initially empty, determine the number of page faults that will occur with the following page replacement algorithms.

- (i) FIFO (First in First out)
- (ii) LRU (Least recently used)

Illustrate the procedure using a suitable diagram.

- (B) **CO-2** Briefly explain different file allocation methods. (8)
BL-2

Q-6. Answer the following. [18]

- (A) **CO-4** Suppose that a disk drive has 300 cylinders, numbered 0 to 299. (8)
BL-5 The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 15. The queue of pending requests, in FIFO order, is 86, 147, 291, 18, 95, 151, 12, 175, and 30. Starting from the current head position, what is the total distance that the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms?

1. First Come First Serve.
2. Shortest Service time First (SSTF)

- (B) **CO-4** Consider the following set of processes with a given CPU burst (10)
BL-5 time and arrival time.

Process Name	Arrival time	Burst Time
A	0	3
B	1	5
C	3	2
D	9	5
E	12	5

Draw the Gantt charts that illustrate the execution of processes and compute the average waiting time and average turnaround time for the following scheduling algorithms:

1. Shortest Process Next (SPN)
2. Round Robin (Time quantum = 2)